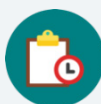


10.5 Comparing Graphical Representations

Lesson Introduction

 Benchmark	MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.		 Learning Target	I can use two graphical representations of data to make comparisons and draw conclusions about the populations.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile and the maximum. Use this summary of the data to describe the spread and distribution of the data. MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.		MA.912.DP.2.1 For two or more sets of numerical univariate data, calculate and compare the appropriate measures of center and measures of variability, accounting for possible effects of outliers. Interpret any notable features of the shape of the data distribution.	
 Essential Questions	- How can you use data displays to compare two data sets? - What measures of center and variation best describe data sets with and without outliers? - How can you use measures of center and variation to draw conclusions about data sets?			
 Lesson Narrative	In this lesson, students apply their knowledge of the measures of center and variation to compare data sets. After a brief Guided Practice, students rotate through three stations comparing data sets presented in line plots, box plots, and stem-and-leaf plots. Students use the data displays to describe the data sets and draw conclusions about the data based on their comparisons.			
 Component Summary	10.5.1 Warm-Up (~5 min.)	Estimate and determine defrosting time using rates (SE p. 129)	10.5.4 Activity: Station 2 (10 min.)	Compare data presented in box plots (SE p. 131)
	10.5.2 Guided Practice (5-10 min.)	Match line plots to data sets based on descriptions (SE p. 129)	10.5.5 Activity: Station 3 (10 min.)	Compare data presented in stem-and-leaf and box plots (SE p. 132)
	10.5.3 Activity: Station 1 (10 min.)	Compare data presented in histograms and line plots (SE p. 130)	10.5.6 Wrap-Up (~5 min.)	Self-reflection on the day's learning target (SE p. 133)
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A	(Optional) Warm-Up Video	- Warm-Up 7.10.5 contains an optional video to accompany the question prompts. - Consider using data from the previous lesson's Ticket out the Door to create small groups ahead of time for the station activities (optional)	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle to draw conclusions from just graphical representations. - Students may struggle to explain the reasons for differences in data for certain contexts. - Students may struggle to recognize that faster means a smaller value in the context of race times.	- Encourage students to consider the features of the data displays, such as shape, spread, and center, to help draw conclusions about the data. - Activate prior knowledge as students work through each scenario to provide additional context for the data. - Consider which groups or stations may need additional support from the teacher and keep this in mind when setting up the flow of each rotation.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns 10.5.4 Activity: Station 2 provides an opportunity to engage students in a Take Turns routine. Students each participate in the work of the activity, then take turns explaining and justifying their responses. This routine encourages all students to participate and prevents 1 or 2 students from monopolizing the mathematical discussion.	MA.K12.MTR.4.1: Discussion and Reflection Each of the Activity Stations engages students in mathematical discourse through partner and small-group work. Encourage students to clarify, critique, and elaborate on their peers' answers to develop students' ability to justify methods and compare their responses to the responses of their peers.
 Differentiate Instruction	10.5.3, 10.5.4, and 10.5.5 Activities provide multiple entry points for students to engage with the content. The graphical representations of the data sets provide visual comparisons that will be particularly helpful for students struggling to compare the data sets. Each activity station is completed through partner or small-group work, providing auditory entry points as students discuss and reason with their peers to complete each activity.	
Lesson Notes:		

10.5.1 Warm-Up: Defrost Time

Component Overview: Play the “Defrost Black Box” video for the class, then release students to work independently to answer the questions.



10.5.1 Warm-Up: Defrost Time

1. A microwave shows that it takes 9 minutes and 45 seconds to defrost 1.5 pounds of meat and 26 minutes to defrost 4 pounds of meat.
 - a. Without performing any calculations, estimate a range of minutes you think it will take to defrost 3.5 pounds of meat.
Answers will vary. It will take approximately 20 to 24 minutes defrost 3.5 pounds of meat.
 - b. Explain how the given information could be used to find the exact amount of time it takes to defrost 3.5 pounds of meat.
Answers will vary. Using the given rates, find the unit rate (time for one pound). Using the unit rate, scale up for the time it would take to defrost 3.5 pounds.
 - c. Determine how long it would take for the microwave to defrost 3.5 pounds of meat. Show your work.
Answers will vary. It will take 22 minutes and 45 seconds to defrost the meat in the microwave.

$$\frac{26}{4} = \frac{6.5}{1} = \frac{22.75}{3.5}$$



This activity provides visual entry points through the presentation of information in the video.

10.5.2 Guided Practice: Comparing Line Plots

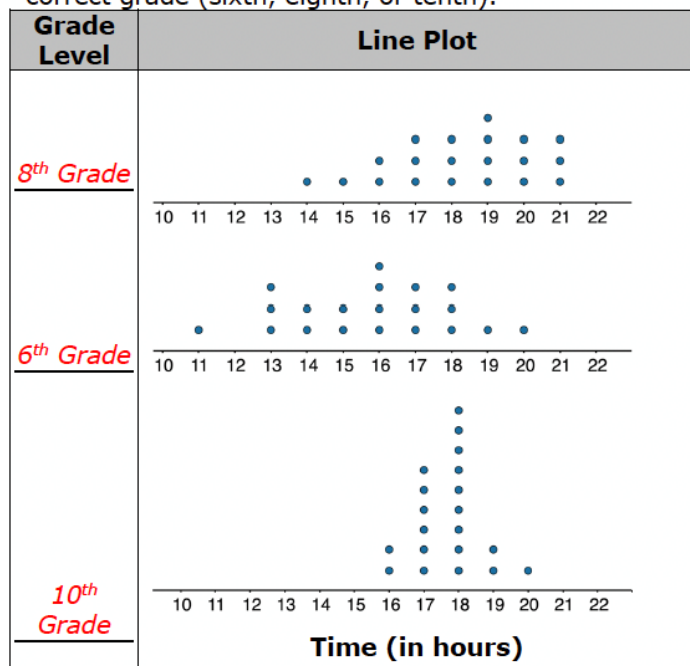
Component Overview: Students match line plots to a data set based on summaries of each set. Students will then rotate through 3 activity stations to continue to apply graphical knowledge of data to make comparisons and draw conclusions about data.



10.5.2 Guided Practice: Comparing Line Plots

- Clare recorded the amount of time a sample of students in the sixth, eighth, and tenth grades spent doing homework, in hours per week. She made a line plot of the data for each grade and provided the following summary:
 - Students in sixth grade tend to spend less time on homework than students in eighth and tenth grades.
 - The homework times for the tenth-grade students are more alike than the homework times for the eighth-grade students.

Use Clare's summary to match each line plot to the correct grade (sixth, eighth, or tenth).



Students completed a similar comparison in Unit 9. The activity reappears as an opportunity to revisit the type of reasoning needed to make the matches and to probe for additional deeper thinking. At this point, students should be able to make additional comparisons between the distributions and discuss more confidently. If, using formative data from prior lessons, you determine that your students do not need this additional support, consider jumping straight into stations following the Warm-Up component.



If students are struggling to make the connection between the line plots and corresponding grade levels, ask prompting questions to provide further scaffolding:

- Where would you expect the 6th grade data to mostly fall on the number line? Why?
- If 10th graders' homework times are "more alike" than 8th graders', does that mean they have more variability or less? How do you know?
- How does a line plot with little variability look compared to one with a lot of variability?



At the conclusion of this component, split students into pairs or small groups and assign each a station to start at. Each group should rotate through all three stations and complete the activities. Allot specific times for each station and review with the class what the signal will be when it is time to transition to the next station.

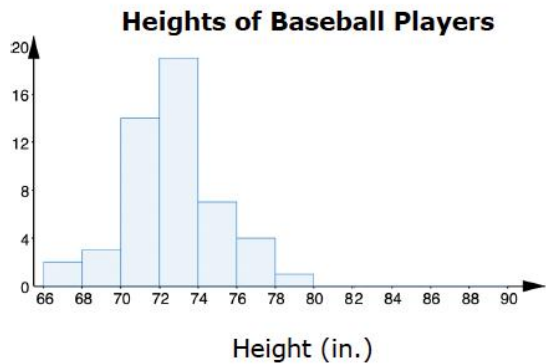
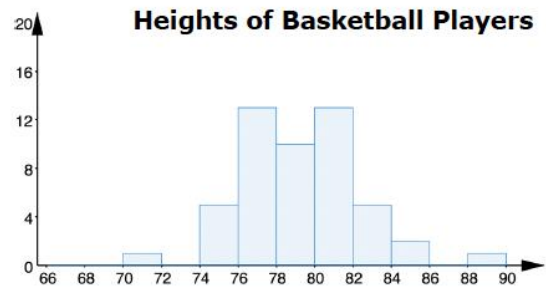
10.5.3 Activity: Station 1 - Histograms and Line Plots

Component Overview: Students compare pairs of data sets presented in histograms and line plots.



10.5.3 Activity: Station 1 – Histograms and Line Plots

- The two histograms show height distributions of 50 male professional baseball players and 50 male professional basketball players, in inches (in.).



This component provides visual entry points through the display of data on the histograms and line plots. Student discussion provides auditory entry points.

10.5.3 Activity: Station 1 - Histograms and Line Plots (continued)

- a. Complete the statements that draw a conclusion about the typical height of professional basketball players and baseball players.

The sample data in the histograms shows the

center of the data for basketball players is

- ☐ less than
☒ greater than

the center of the data for baseball

players. It can be concluded that professional

basketball players are typically

- ☐ shorter than
☒ taller than

professional baseball players.

- b. Complete the statement.

The histograms show that in the samples, the

range of heights of basketball players is

- ☐ less than
☒ greater than

the range of the heights of

baseball players. We can conclude that professional

basketball players have

- ☒ more
☐ less

variability in

their heights than professional baseball players.

One possible reason for this is ...

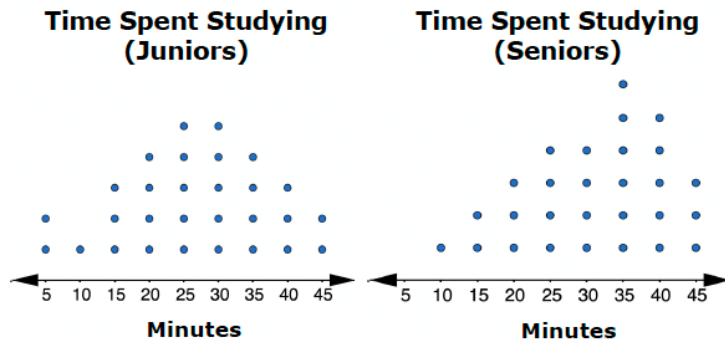
Answers will vary. A possible reason for the variation could be the different heights have benefits for different positions on a basketball team.



Question 1b challenges students to consider why there is a difference in the heights of basketball players vs. baseball players. Students who may not watch or know a lot about sports may struggle to answer this question because they are not familiar with the context. Have group members discuss what they know about the skill requirements for each sport and how that might impact the populations/data.

10.5.3 Activity: Station 1 - Histograms and Line Plots (continued)

2. Mr. Logan surveyed a sample of his junior and senior students about the time they spend studying math per day. He then tabulated the results and created a line plot displaying the data for each group.



- The value of the larger median study time for the two groups is 32.5.
- The value of the larger mean study time for the two groups is 30.9.
- In one to two sentences, describe the difference between the number of minutes Mr. Logan's juniors and seniors study by comparing the center and shapes for the groups.

Answers will vary. According to the mean of both groups, seniors study an average of 4.3 minutes longer than juniors. The median study time for seniors is 7.5 minutes higher than the median for juniors. The juniors' line plot appears more symmetrical than the seniors, so there are some seniors who are pulling the mean hours studying above the median.



As students work through this component in their small groups, circulate and ask guiding questions to prompt their thinking:

- What do you notice about the spread of the data on each line plot? What does this tell you about the time each group spends studying?
- Without doing the calculations, which measures of center do you think are best to describe each data set? Why?

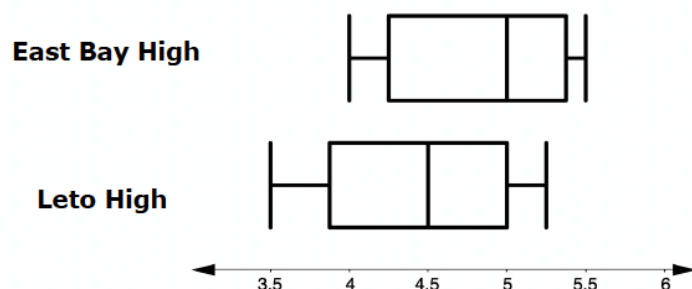
10.5.4 Activity: Station 2 - Box Plots

Component Overview: At Station 2, students compare data presented in two box plots and describe features of the data sets based on the box plots. Students then draw conclusions based on the presented data and what they know about measures of center and variability and conclude by creating a summary describing what they know about the data sets.



10.5.4 Activity: Station 2 – Box Plots

- The mile times for a sample of track team members at East Bay High School and Leto High School are shown in the two box plots.



Mile Time (minutes)

- Using the box plots above, calculate and describe each of the features for Leto High in the table.

Feature	East Bay High School	Leto High School
Shape	The shape of the data is asymmetrical.	<i>Answers will vary. The shape of the data is asymmetrical.</i>
Center	The median mile time for the sample of East Bay High team members is 5 minutes.	<i>The median mile time for the sample of Leto High is 4.5 minutes.</i>
Variability	The range of the sample of East Bay High team members' mile times is 1.5 minutes. The IQR of the sample of East Bay High team members' mile times is 0.75 minutes.	<i>The range of the sample of Leto High team members' mile times is 1.75 minutes. The IQR of the sample of Leto High team members' mile times is 1.2 minutes.</i>



Take Turns

Consider implementing a Take Turns routine during this station. Have pairs or groups work to complete each part of this activity, then take turns justifying their responses to their peers. Encourage students to explain whether they agree or disagree with each other's' arguments and why.

10.5.4 Activity: Station 2 - Box Plots (continued)

- b. Based on the data, explain which team is more likely to win at a track meet.
Answers will vary. Leto High would be more likely to win at a track meet. Their median time is faster than East Bay.
- c. Suppose a runner at a track meet runs a mile in 3 minutes and 45 seconds. Explain which track team you would assume they were on, using information from the data displays.
Answers will vary. The runner is most likely a member of the Leto High team because their data supports a runner finishing under 4 minutes.
- d. Based on your conclusions and answers to Parts A through C, use some of the terms from the word bank below to write a 4-5 sentence summary comparing the population of runners at East Bay High School to the population of runners at Leto High School.

mean	median	range	IQR
center	variability	upper quartile	
lower quartile	sample	population	

Answers will vary. Leto High School has faster runners than East Bay High. The median time for Leto High is 4.5 minutes where East Bay has a median of 5 minutes. On average, Leto High is finishing half a minute faster than East Bay. East Bay High's spread is smaller than Leto High, but they have times longer than Leto High.



Students may struggle to recognize that faster times mean a shorter completion time. For example, some students may think East Bay High is faster because their median is higher.



As students work through this station, ask questions to encourage synthesis of the material:

- *How can you use the box plots to determine which team is more likely to win the track meet?*
- *Which team is faster? How do you know? How can you use this to help you answer the questions and summarize the data?*



Challenge students who may finish early to make predictions about how different times might impact the spread of the data sets. What if the runner who ran a mile in 3 minute and 45 seconds was a member of the East Bay High School team? Would this significantly impact the data? Why or why not?



Students may have a hard time drawing conclusions about the populations to compare the data. Remind them to compare shape, center, and variability. Ensure students understand that the data sets are generally representative of the populations and can therefore be predictive of the populations at each school.

10.5.5 Activity: Station 3 - Stem-and-Leaf Plots and Box Plots

Component Overview: At Station 3, students analyze the data of two groups displayed in stem-and-leaf plots and two groups displayed in box plots. Students draw conclusions about the data sets and apply their knowledge of measures of center and variability to make determinations about the data.



10.5.5 Activity: Station 3 – Stem-and-Leaf Plots and Box Plots

1. A sample of two stores' guitar prices are shown in the stem-and-leaf plots.

Store A's Guitar Prices

Stem	Leaf
2	1, 2, 8
3	3, 6
4	5, 8, 8
5	1, 6
6	
7	2, 2
8	1, 6
9	1

Key: 2|1 = \$210.00

Store B's Guitar Prices

Stem	Leaf
4	5, 8
5	6, 9
6	1, 1, 3, 5
7	5, 5, 6
8	0, 6
9	1, 8
10	0

Key: 2|1 = \$210.00

Zach wants to purchase a guitar and is trying to determine which store to purchase from. Using the displays he makes the following conclusion.

"Store B is better for more experienced guitar players because it typically sells more expensive guitars than Store A. The median price of Store B's guitars is \$700, while the median price of Store A's guitars is \$480. Prices of guitars at Store A tend to vary more, with an IQR of \$390, compared to the prices of guitars at Store B, with a much smaller IQR."

Do you agree or disagree with Zach's conclusion? Explain your reasoning.

Answers will vary. The calculations Zach has made are correct. However, those calculations alone cannot determine if Store B is for more experienced guitar players. Store B could just carry higher quality or specific name brands of guitars whereas Store A may sell a variety of name brands and quality.

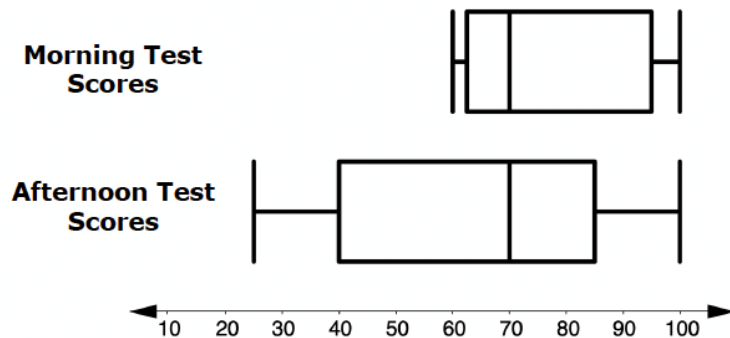


Think-Pair-Share

Engage students in a Think-Pair-Share routine as they determine whether they agree or disagree with Zach's conclusion. First, students analyze the stem-and-leaf plots and consider whether they agree with Zach's response. Next, students share their thoughts with a partner. Encourage students to explain whether they agree or disagree with their partner and why. Finally, students complete their written response that may later be shared with the class.

10.5.5 Activity: Station 3 - Stem-and-Leaf Plots and Box Plots (continued)

2. Suppose a sample of test scores is taken from Mr. Spellman's morning math classes. The data is displayed in the top box plot. Another sample of test scores is taken from Mr. Spellman's afternoon math classes. The data is displayed in the bottom box plot.



Explain the difference between taking Mr. Spellman's math test in the morning compared to the afternoon. Use some of the terms from the word bank and be sure to include both a measure of center and variability in your explanation.

mean	median	range	IQR
center	variability	upper quartile	
lower quartile	sample	population	

Answers will vary. The mean cannot be determined from the box plots above. The median would be the best measure of center. Both the morning and afternoon students had a median score of 70. The variation in scores for the morning students is lower than the afternoon students. The range for the morning students was 40 whereas the range for the afternoon students was 75. There are a variety of factors that could cause the spread in data to be different such as students did not study or they were more alert in the morning as compared to the afternoon.



As students complete this component, ask questions to solidify understanding:

- Which measures of center/variability are best to use for the data in the box plots? Why?
- How do the representations look different?
- What do you notice about the box plots just by looking at them?
- What conclusions can you make about the samples?

10.5.6 Wrap-Up: Reflection

Component Overview: Students complete a reflection to determine what they are still unsure of after the lesson.



10.5.6 Wrap-Up: Reflection

Complete each statement based on your current understanding.

1. I am still unsure about Answers will vary.
2. One question I have that will help me better understand is Answers will vary.



For additional data points, consider using student responses from the Activities Stations.